

Supplementary File

Figure S1

Figure S1(supplement): Data flow from overall study population to matched dataset; DBD, determination of death by neurological criteria; DCD, determination of death after cardiocirculatory arrest; MELD, Model of End Stage Liver Disease; SCS, static cold storage; NMP, normothermic machine perfusion

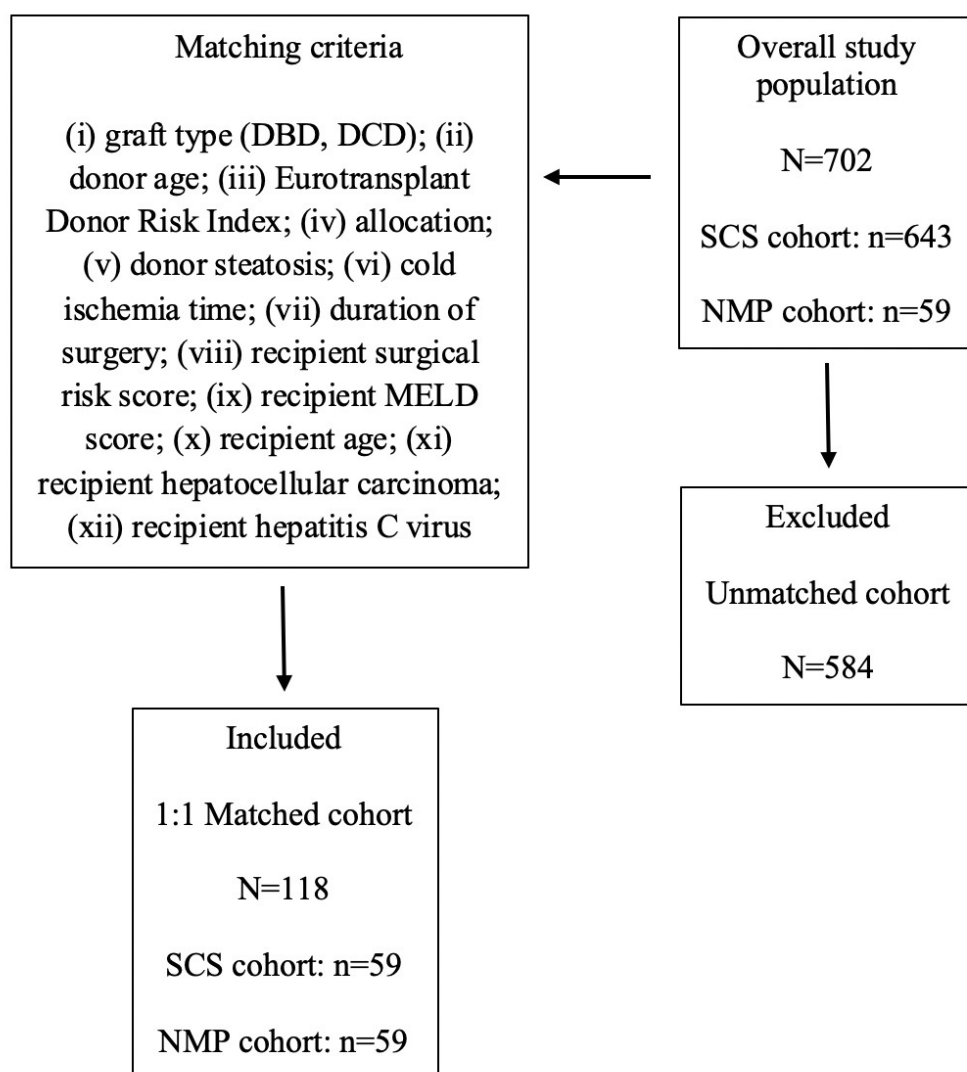


Figure S2

Figure S2 (supplement): Propensity score distributions by treatment groups: SCS (n=59) distance (mean)=0.24 (SD: 0.17, range: 0.00-0.77); NMP (n=59) distance mean=0.32 (SD: 0.28, range: 0.00-0.96); SCS, static cold storage; NMP, normothermic machine perfusion

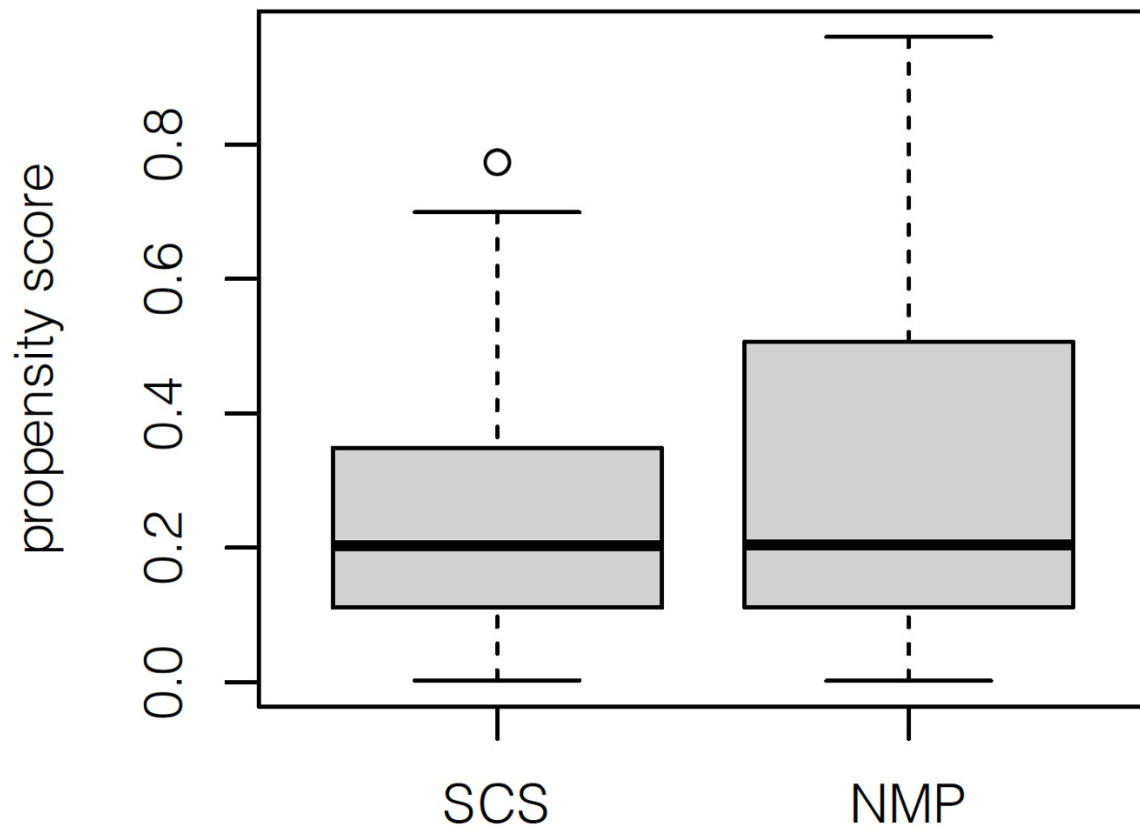


Figure S3

Figure S3 (supplement): Indications for NMP in transplanted (a) and not transplanted (b) livers

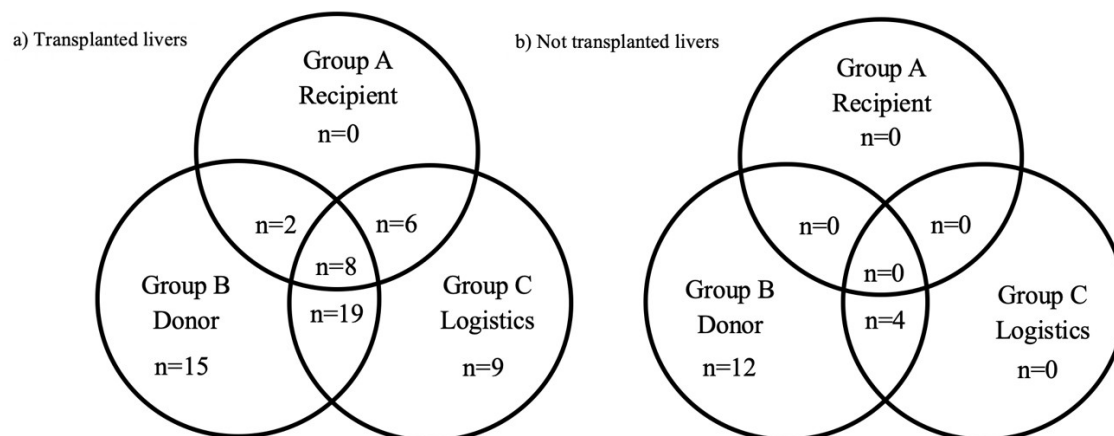


Table S1: Baseline Characteristics of the Study Population

Item	Total	SCS	NMP	P value
(N)	761	702	59	
Eurotransplant Donor Risk Index*	2 (0)	2 (0)	2 (1)	0.008
Donor Age (Years)*	52.0 (22.0)	52.0 (23.0)	57.0 (22.0)	0.074
Donation After Cardiac Death				p<.001
Yes	16 (2)	7 (<1)	9 (15)	
No	745 (98)	695 (99)	50 (85)	
Donor Steatosis				0.142
none	428 (56)	400 (57)	28 (47)	

Item	Total	SCS	NMP	P value
mild	249 (33)	222 (32)	27 (46)	
medium	71 (9)	68 (10)	3 (5)	
severe	13 (2)	12 (2)	1 (2)	
Cold Ischemia Time (h)*	8.00 (3.00)	8.50 (3.00)	6.00 (3.00)	p<.001
Allocation				0.006
local	269 (35)	259 (37)	10 (17)	
regional	327 (43)	297 (42)	30 (51)	
national	165 (22)	146 (21)	19 (32)	
Surgery Duration (Min)*	388 (140)	387 (140)	388 (128)	0.570
Surgical Risk Score				p<.001
none	1 (<1)	.	1 (2)	
low	597 (78)	571 (81)	26 (44)	
middle	55 (7)	44 (6)	11 (19)	
high	108 (14)	87 (12)	21 (36)	
Recipient Age (Years)*	58.9 (12.3)	58.6 (12.7)	63.4 (12.7)	0.008
MELD Score*	18 (14)	18 (15)	15 (12)	0.015
Recipient Hepatocellular Carcinoma				0.578
Yes	244 (32)	227 (32)	17 (29)	
No	517 (68)	475 (68)	42 (71)	
Recipient Hepatitis C Virus				0.579

Item	Total	SCS	NMP	P value
negative	709 (93)	653 (93)	56 (95)	
positive	52 (7)	49 (7)	3 (5)	

*Values in parentheses are percentages unless indicated otherwise; *values are median (i.q.r.); SCS, static cold storage; NMP, normothermic machine perfusion; MELD, Model of End Stage Liver Disease*

Chi-square and Fisher's Exact Test for categorical variables and Mann-Whitney-U-Wilcoxon Test for continuous variables

Table S2: Matched Data 1:1

Item	Total	SCS	NMP	P value
(N)	118	59	59	
Eurotransplant Donor Risk Index*	2 (1)	2 (0)	2 (1)	0.951
Donor Age (Years)*	56.0 (20.0)	56.0 (16.5)	57.0 (22.0)	0.808
Donation After Cardiac Death				0.142
Yes	13 (11)	4 (7)	9 (15)	
No	105 (89)	55 (93)	50 (85)	
Donor Steatosis				0.587
none	61 (52)	33 (56)	28 (47)	
mild	47 (40)	20 (34)	27 (46)	
medium	7 (6)	4 (7)	3 (5)	
severe	3 (3)	2 (3)	1 (2)	
Cold Ischemia Time (h)*	7 (3)	7 (3)	6 (12)	0.061
Allocation				0.822
local	18 (15)	8 (14)	10 (17)	
regional	63 (53)	33 (56)	30 (51)	
national	37 (31)	18 (31)	19 (32)	
Surgery Duration (Min)*	378 (129)	366 (126)	388 (128)	0.460
Surgical Risk Score				0.439
none	1 (<1)	.	1 (2)	
low	49 (42)	24 (41)	25 (42)	
middle	28 (24)	17 (29)	11 (19)	
high	40 (34)	18 (31)	22 (37)	

Item	Total	SCS	NMP	P value
Recipient Age (Years)*	61.2 (11.9)	60.6 (10.5)	63.4 (12.7)	0.260
MELD Score*	16 (12)	17 (10)	15 (12)	0.478
Recipient Hepatocellular Carcinoma				0.689
Yes	36 (31)	17 (29)	19 (32)	
No	82 (69)	42 (71)	40 (68)	
Recipient Hepatitis C Virus				0.697
negative	111 (94)	55 (93)	56 (95)	
positive	7 (6)	4 (7)	3 (5)	

*Values in parentheses are percentages unless indicated otherwise; *values are median (i.q.r.); SCS, static cold storage; NMP, normothermic machine perfusion; MELD, Model of End Stage Liver Disease*

Chi-square and Fisher's Exact Test for categorical variables and Mann-Whitney-U-Wilcoxon Test for continuous variables

Table S3: Indications for NMP and result in transplanted/discarded livers

Indication for NMP	Total NMP patients	Result in transplan t	Result in discard
A	0	0	0
B	27	15	12
C	9	9	0
A & B	2	2	0
A & C	6	6	0
B & C	23	19	4
A & B & C	8	8	0
Total	75	59	16

NMP, normothermic machine perfusion; A= recipient; B= donor; C= logistics

Table S4: Severity scoring system for Ischemic Type Biliary Lesions (ITBL) classification:
mild (1-2) moderate (3-4), severe (5-6)

Morphological features	Localization	Score
Strictures		
	Extrahepatic common bile duct, including the hilar bifurcation	1
	Unilateral hepatic duct, second- and third order bile ducts	2
	Bilateral hepatic ducts, second- and third order bile ducts	3
Ductal dilatation		
	Intrahepatic unilateral bile ducts	2
	Intrahepatic bilateral bile ducts	3

Subsample Analysis: NMP for donor-predominant indications

Table S5: Subsample analysis NMP for donor-predominant indications: recipient characteristics

Item	SCS		NMP		P value
	n	m	n	m	
Recipient Age (Years)*	42	59.02 (7.62)	42	59.83 (11.62)	0.164
Body Mass Index (kg/m ²)*	42	25.42 (4.40)	42	24.58 (4.55)	0.622
Recipient Sex	42		42		01.000
male		36 (86)		36 (86)	
female		6 (14)		6 (14)	
MELD Score*	42	18.29 (9.37)	42	19.36 (9.26)	0.733
Indication for Liver Transplantation					
Alcoholic Steatohepatitis	42	7 (17)	42	9 (21)	0.578
Non-alcoholic Steatohepatitis	42	5 (12)	42	7 (17)	0.533
Primary non-function	42	1 (2)	42	.	0.314
Polycystic liver disease	42	1 (2)	42	.	0.314
Budd Chiari Syndrome	42	.	42	1 (2)	0.314
Primary Biliary Cirrhosis	42	1 (2)	42	1 (2)	01.000
Primary Sclerosing Cholangitis	42	.	42	3 (7)	0.078
Cryptogenic Liver Cirrhosis	42	5 (12)	42	.	0.021
Hepatitis Virus Infection	42	2 (5)	42	3 (7)	0.645
Cholangitic Abscess Formation	42	6 (14)	42	2 (5)	0.137
Alpha 1-antitrypsin Deficiency	42	1 (2)	42	3 (7)	0.306
Tumor	42	16 (38)	42	17 (40)	0.823
Tumor.Entity					
Hepatocellular Carcinoma	42	12 (29)	42	16 (38)	0.355
Cholangiocellular Carcinoma	35	3 (9)	40	.	
Liver metastases of colon malignancy	35	.	40	1 (2)	
Other	35	1 (3)	40	.	
Tumor Treatment	36	12 (33)	41	14 (34)	0.940
Transarterial Chemoembolization	36	7 (19)	41	8 (20)	0.994
Radiofrequency Ablation	36	6 (17)	41	8 (20)	0.747

Item	SCS		NMP		P value
	n	m	n	m	
CHILD Score	34		32		0.187
A	20 (29)		10 (31)		
B	20 (59)		13 (41)		
C	4 (12)		9 (28)		
AB0 blood group	42		42		0.177
A	19 (45)		20 (48)		
B	4 (10)		8 (19)		
0	13 (31)		13 (31)		
AB	6 (14)		1 (2)		
Hepatitis C Virus	42		41		0.980
negative	40 (95)		39 (95)		
positive	2 (5)		2 (5)		
Encephalopathy	42	10 (24)	42	6 (14)	0.266
Cirrhosis	42	32 (76)	42	37 (88)	0.154
Surgical Risk Score	42		42		0.916
none	.		.		
low	20 (48)		20 (48)		
middle	9 (21)		10 (24)		
high	13 (31)		12 (29)		
Portal Vein Open	42		39		0.411
Yes	38 (90)		32 (82)		
Partial obstruction of portal vein trunk	2 (5)		1 (3)		
Complete obstruction of portal vein trunk	1 (2)		4 (10)		
Thrombosis right or left portal branch	1 (2)		2 (5)		

*Values in parentheses are percentages unless indicated otherwise; *values are median (i.q.r.); SCS, static cold storage; NMP, normothermic machine perfusion; MELD, Model of End Stage Liver Disease*

Chi-square and Fisher's Exact Test for categorical variables and Mann-Whitney-U-Wilcoxon Test for continuous variables

Table S6: Subsample analysis NMP for donor-predominant indications: donor characteristics and operative data

Item	SCS		NMP		P value
	n	m	n	m	
Donor Age (Years)*	42	56.5 (15.8)	42	60.0 (20.0)	0.171
Donor Sex	42		42		0.659
male		25 (60)		23 (55)	
female		17 (40)		19 (45)	
Donor Body Mass Index (kg/m ²)*	42	25.70 (4.12)	42	26.60 (3.30)	0.369
Donor Height (m)*	42	1.8 (0.1)	42	1.7 (0.2)	0.415
Donor Weight (kg)*	42	77.0 (13.8)	42	80.0 (20.0)	0.735
Steatosis	42		41		0.667
none		20 (48)		16 (39)	
mild		17 (40)		22 (54)	
medium		4 (10)		3 (7)	
severe		1 (2)		.	
Eurotransplant Donor Risk Index*	42	2 (0)	41	2 (1)	0.179
Donor Artery Anatomy	42		40		0.381
Normal		32 (76)		27 (68)	
Variance		10 (24)		13 (32)	
Donation After Cardiac Death	42	4 (10)	41	9 (22)	0.119
Allocation	42		42		0.668
local		3 (7)		5 (12)	
regional		24 (57)		23 (57)	
national		15 (36)		12 (30)	
Cause of Death	42		40		0.614
Trauma		5 (12)		6 (15)	
Anoxia		6 (14)		9 (22)	
Cardiovascular Event		29 (69)		22 (55)	
Other		2 (5)		3 (8)	

Item	SCS		NMP		P value
	n	m	n	m	
Intensive Care Unit Length Of Stay (Days)*	3	(4)	40	4 (4)	0.911
Sodium (U/L)*	42	146 (8)	37	144 (9)	0.619
Latest GGT*	42	38 (73)	40	75 (94)	0.077
Simultaneous Transplant	40		40		01.000
No		40 (100)		40 (100)	
Liver-Kidney		.		.	
Heart-Liver		.		.	
Multi-Organ		.		.	
Surgery Duration (Min)*	42	346 (108)	40	381 (138)	0.528
Mass Clamping Lig. Hepatoduodenale	42	.	40	1 (2)	0.303
Anastomosis Time (Min)*	42	42 (10)	40	50 (10)	0.003
Cold Ischemia Time (h)*	42	7 (2)	40	6 (2)	0.163
Total Preservation Time (h)*	42	7 (2)	41	24 (12)	p<.001

*Values in parentheses are percentages unless indicated otherwise; *values are median (i.q.r.);*

SCS, static cold storage; NMP, normothermic machine perfusion; GGT, gamma-glutamyl transferase

Chi-square and Fisher's Exact Test for categorical variables and Mann-Whitney-U-Wilcoxon Test for continuous variables

Table S7: Subsample analysis NMP for donor-predominant indications: clinical outcomes and complications

Item	SCS		NMP		P value
	n	m	n	m	
Lenght Of Stay (Days)*	42	24 (22)	38	16 (12)	0.003
Clavien Dindo Classification	42		41		0.097
I		2 (5)		.	
II		18 (43)		11 (27)	
IIIa		.		5 (12)	
IIIb		7 (17)		6 (15)	
IVa		11 (26)		13 (32)	
V		4 (10)		6 (15)	
Early Allograft Dysfunction	41	16 (39)	42	15 (36)	0.755
Vascular Complications	42	9 (21)	42	5 (12)	0.242
Arterial Complications	42	5 (12)	42	2 (5)	0.236
Arterial Dissection	42	1 (2)	42	1 (2)	01.000
Arterial Thrombosis	42	4 (10)	42	.	0.040
Arterial Stenosis	42	2 (5)	42	1 (2)	0.557
Portal Vein Stenosis	42	1 (2)	42	1 (2)	01.000
Venous Thrombosis	42	5 (12)	42	3 (7)	0.457
Re-Operation	42	13 (31)	42	13 (31)	=1.000
Re-Operation $\leq$ 30 Days	42	12 (29)	42	14 (33)	0.637
Haematoma	42	7 (17)	42	9 (21)	0.578
Acute Kidney Failure	42	7 (17)	42	5 (12)	0.533
Bile Duct Complications	42	20 (48)	42	19 (45)	0.827
Bile Duct Complications $\leq$ 30 days	42	9 (21)	42	8 (19)	0.786
Bile Duct Complications 30 days	42	14 (33)	42	11 (26)	0.474
Bile Duct Leak	42	8 (19)	42	5 (12)	0.365
Anastomotic Strictures	42	15 (36)	42	15 (36)	01.000
Non Anastomotic Strictures	42	7 (17)	42	4 (10)	0.332
Ischemic Type Biliary Lesions (ITBL)	42	6 (14)	42	2 (5)	0.137

Item	SCS		NMP		P value
	n	m	n	m	
ITBL grade	6		2		n.a.
Mild	1 (17)		2 (100)		
Moderate	1 (17)		.		
Severe	4 (67)		2 .		
Intrahepatic Cast	17 1 (6)		14 .		0.356
Extrahepatic Cast	17 4 (24)		14 .		0.052
Cholangitis	42 7 (17)		42 3 (7)		0.178

*Values in parentheses are percentages unless indicated otherwise; *values are median (i.q.r.); SCS, static cold storage; NMP, normothermic machine perfusion*

Chi-square and Fisher's Exact Test for categorical variables and Mann-Whitney-U-Wilcoxon Test for continuous variables

Table S8: Subsample analysis NMP for donor-predominant indications: patient and graft survival in days

			n (per cent [95 per cent CI])		
		n	30	90	365
Patient	NMP	42	2(95[89,100])	3(86[76,98])	0 (86[76,98])
	SCS	42	1(98[93,100])	2(93[85,100])	5(80[68,93])
Log Rank-Test $\chi^2_{(1)}=0.01$, 0.942					
Graft	NMP	42	2(95[89,100])	3(86[76,98])	0(86[76,98])
	SCS	42	3(93[85,100])	1(90[82,100])	6(75[62,90])
Log Rank-Test $\chi^2_{(1)}=0.30$, 0.586					

SCS, static cold storage; NMP, normothermic machine perfusion

Subsample Analysis: ITBL patients

Table S9: Subsample analysis for ITBL patients: recipient characteristics

Item	SCS		NMP	
	n	m	n	m
Recipient characteristics				
Recipient Age (Years)*	8	63.7 (9.4)	2	58.8
Body Mass Index (kg/m ²)*	8	25.35 (1.20)	2	22.45
Recipient Sex	8		2	
male		7 (88)		1 (50)
female		1 (12)		1 (50)
MELD Score*	8	16 (19)	2	16
Indication for Liver Transplantation				
Alcoholic Steatohepatitis	8	1 (12)	2	1 (50)
Non-alcoholic Steatohepatitis		.		.
Primary non function		.		.
Polycystic liver disease		.		.
Budd Chiari Syndrome	8	.	2	1 (50)
Primary Biliary Cirrhosis		.		.
Primary Sclerosing Cholangitis		.		.
Cryptogenic Liver Cirrhosis		.		.

Item	SCS		NMP	
	n	m	n	m
Hepatitis Virus Infection	.		.	
Cholangitic Abscess Formation	8	1 (12)	.	
Alpha 1-antitrypsin Deficiency	.		.	
Acute Liver Failure	8	1 (12)	2	.
Tumor	8	6 (75)	2	.
Tumor Entity				
Hepatocellular Carcinoma	8	6 (75)	2	.
Cholangiocellular Carcinoma	.		.	
Neuroendocrine Tumor	.		.	
Liver metastases of colon malignancy	.		.	
Other	.		.	
Tumor Treatment	6		2	
Yes	4	(67)	.	
No	2	(33)	2	(100)
Transarterial Chemoembolization	6		2	
Yes	2	(33)	.	
No	4	(67)	2	(100)
Radiofrequency Ablation	6		2	
Yes	4	(67)	.	
No	2	(33)	2	(100)
CHILD Score	7		1	
A	5	(71)	.	
B	1	(14)	.	
C	1	(14)	1	(100)
AB0 blood group	7		2	
A	3	(43)	.	
B	1	(14)	1	(50)
0	2	(29)	1	(50)
AB	1	(14)	.	
Hepatitis C Virus	8		2	

Item	SCS		NMP	
	n	m	n	m
negative	7	(88)	2	(100)
positive	1	(12)	.	
Encephalopathy	8		2	
Yes	2	(25)	1	(50)
No	6	(75)	1	(50)
Cirrhosis	8		2	
Yes	6	(75)	1	(50)
No	2	(25)	1	(50)
Surgical Risk Score	8		2	
none	.		.	
low	2	(25)	.	
middle	4	(50)	2	(100)
high	2	(25)	.	
Portal Vein Open	8		1	
Yes	8	(100)	1	(100)
Partial obstruction of portal vein trunk	.		.	
Complete obstruction of portal vein trunk	.		.	
Thrombosis right or left portal branch	.		.	

*Values in parentheses are percentages unless indicated otherwise; *values are median (i.q.r.); SCS, static cold storage; NMP, normothermic machine perfusion; MELD, Model of End Stage Liver Disease*

Chi-square and Fisher's Exact Test for categorical variables and Mann-Whitney-U-Wilcoxon Test for continuous variables

Table S10: Subsample analysis for ITBL patients: donor characteristics and operative data

Item	SCS		NMP	
	n	m	n	m
Donor Age (Years)*	8	58.0 (11.8)	2	62.5
Donor Sex	8		2	
male		4 (50)		.
female		4 (50)		2 (100)
Donor Body Mass Index (kg/m ²)*	8	25.45 (3.4)	2	23.25
Donor Height (m)*	8	1.7 (0.1)	2	1.7
Donor Weight (kg)*	8	78.0 (14.2)	2	65.0
Donor after Cardiac Death	8		2	
Yes		1 (12)		.
No		7 (88)		2 (100)
Steatosis	8		2	
none		5 (62)		1 (50)
mild		2 (25)		1 (50)
medium		1 (12)		.
severe		.		.
Eurotransplant Donor Risk Index*	8	2 (0)	2	2
Donor Artery Anatomy	8		2	
Normal		5 (62)		.
Variance		3 (38)		2 (100)
Allocation	8		2	
local		.		.
regional		6 (75)		2 (100)
national		2 (25)		.
Cause of Death	8		2	
Trauma		3 (38)		.
Anoxia		1 (12)		1 (50)
Cardiovascular Event		3 (38)		1 (50)

Item	SCS		NMP	
	n	m	n	m
Other		1 (12)		.
Intensive Care Unit Length Of Stay (Days)*	8	2 (1)	2	4
Sodium (U/L)*	7	147 (6)	1	143
Latest GGT*	8	64 (62)	2	58
Simultaneous Transplant	8		2	
No		8 (100)		2 (100)
Liver-Kidney		.		.
Heart-Liver		.		.
Multi-Organ		.		.
Surgery Duration (Min)*	8	358 (86)	2	314
Mass Clamping Lig. Hepatoduodenale	8		2	
Yes		.		.
No		8 (100)		2 (100)
Anastomosis Time (Min)*	8	45 (14)	2	44
Cold Ischemia Time (h)*	8	7 (2)	2	7
Total preservation time (h)*	8	7 (2)	2	22

*Values in parentheses are percentages unless indicated otherwise; *values are median (i.q.r.);*

SCS, static cold storage; NMP, normothermic machine perfusion; GGT, gamma-glutamyl transferase

Chi-square and Fisher's Exact Test for categorical variables and Mann-Whitney-U-Wilcoxon Test for continuous variables

Table S11: Costs of ICU and overall length of stay per patient

	SCS	NMP
ICU-Stay	median 4,63 days	median 4,65 days
Costs ICU	19.588,6€/ patient	19.673,2€/patient
Length of Stay	median 33.27 days	median 24.4 days
Cost overall stay/day	92.317,6€/patient	67.705,12€/patient

ICU: Intensive care unit; SCS: Static cold storage; NMP: Normothermic machine perfusion

Table S12: Costs of ERCP per ITBL-patient

	ERCP
SCS/ITBL (n=8)	1.329,2€
NMP/ITBL (n=2)	332,3€

ERCP, Endoscopic retrograde cholangiopancreatography; ITBL, Ischemic Type Biliary Lesions; SCS: static cold storage; NMP: normothermic machine perfusion